

[DC] Supporting Asynchronous Collaboration in Mixed Reality with Large Language Models (LLMs)

Giuseppina Pinky Kathlea Diatmiko*
IMT Atlantique, Lab-STICC, UMR CNRS 6285, France

ABSTRACT

While prior work has demonstrated the benefits of asynchronous collaboration in Mixed Reality (MR) for tasks like education and industrial training, it is still under-investigated compared to synchronous collaboration. Previous asynchronous MR systems rely mainly on recording and replaying of the collaborators' actions, which raises many challenges in terms of interactivity, understanding and processing speed of the recorded data. My PhD project investigates how Large Language Models (LLMs) can be used to improve asynchronous collaboration by creating a virtual surrogate which can interact on behalf of a person who is not available at a particular time. This paper presents the research approach, the application context and the first prototype developed as part of this project, as well as the future work planned for the upcoming years.

Index Terms: Mixed reality, virtual reality, asynchronous collaboration, large language models, AI-agent, collaborative design.

1 INTRODUCTION

Advances in digital technology make it possible to foster collaborative work between people who are separated in space and time. In this context, *asynchronous collaboration* is extremely common, as when people exchange e-mails, leave notes in shared documents, or send voice messages. This type of collaboration enables multiple collaborators to work across different time zones or working hours. It also offers many advantages, such as allowing work parallelism, avoiding interruptions, and providing additional reflection time.

Prior work on Mixed Reality (MR), including Virtual Reality (VR) and Augmented Reality (AR), has extensively investigated *synchronous collaboration*, in which several people interact in real-time in a shared environment or with shared 3D content, as demonstrated by recent surveys [8, 4]. However, *asynchronous collaboration* in MR has been scarcely investigated, as reported by Irlitti et al. [5]. For example, a recent survey by Sereno et al. [9] on collaborative AR listed 3 articles on asynchronous collaboration compared to 47 on synchronous collaboration. Nevertheless, asynchronous collaboration in MR is crucial, as it enables collaborators to work on immersive tasks at different time. It also offers several benefits over synchronous collaboration in this context. For instance, the system proposed by Fender and Holz [3] uses asynchronous collaboration to avoid interruptions between collaborators. In a follow-up study [6], they have shown that participants felt less frustrated and stressed with this system compared to synchronous collaboration. Previous work has already provided useful use cases for asynchronous MR collaboration in various application domains, including education [10, 11] and industry [1, 7].

Previous asynchronous MR systems [1, 3, 7, 10, 11] rely mainly on two steps: a recording step of one collaborator's actions and a playback step by another collaborator. However, this approach presents many challenges in terms of interactivity and efficiency

when exploring the recorded data and understanding the actions taken, making it difficult to support a truly iterative process between collaborators. My PhD project investigates how Large Language Models (LLMs) as virtual surrogates can be used to address these challenges and improve asynchronous collaboration in MR. In particular, it aims to focus on the following research questions:

1. Can an LLM understand context and capture knowledge during the recording step?
2. Can an LLM be used to support interactivity during the playback step, while shortening review of the recorded data?
3. Can an LLM help support a truly iterative process between collaborators working with multiple iterations over time?

To address these questions, I plan to explore collaborative design scenarios in which multi-disciplinary experts have to go through several iterative steps before reaching the final design.

2 RELATED WORK

Previous research on MR asynchronous collaboration has mainly concentrated on the concept of sharing pre-recorded activities for later use. This approach has found applications across various domains. In education, XR-Live [10] enabled remote students to engage with laboratory activities by observing and replaying pre-recorded instructor demonstrations within a virtual environment. Similarly, Virtual Triplets [11] allowed students in a cooking class to review recorded demonstrations by instructors, facilitating asynchronous learning.

The utility of recording and replaying actions in VR has also been explored in professional contexts. The AsyncReality system [3], for example, supports the playback of recorded physical events in an office setting, enabling users to stay immersed in their tasks while remaining informed about missed activities. Asynchronous collaboration is particularly prevalent in object assembly tasks within MR. Cho et al. [1] developed a system that provides playback of user activities in MR, specifically for assembly training purposes. Similarly, Mayer et al. [7] investigated asynchronous collaboration in remote assembly training and found that participants using recording and replay features experienced reduced cognitive load compared to those relying on traditional manuals or instructional text in VR. These examples highlight the evolving role of asynchronous collaboration in facilitating effective knowledge transfer and task completion across diverse application domains.

Asynchronous collaboration in MR presents unique challenges, particularly due to the absence of real-time verbal communication. This lack of immediate interaction can limit the richness of information exchange and hinder the collaborative process, as reported by Chow et al. [2]. Unlike synchronous collaboration, asynchronous approaches rely heavily on understanding recorded object manipulation in a virtual environment without additional information. This can lead to potential misinterpretations or inefficiencies in translating design intent or collaborative objectives. Therefore, integrating asynchronous collaboration applications with complementary tools to ensure the continuity and coherence of collaborative processes warrants further investigation.

*e-mail: giuseppina-pinky.diatmiko@imt-atlantique.fr

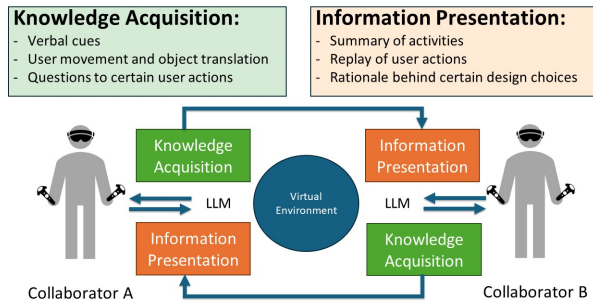


Figure 1: LLM-supported asynchronous collaboration in MR.

3 LLMs TO SUPPORT ASYNCHRONOUS COLLABORATION

Recent advances in generative AI using LLMs demonstrate natural and live interaction with a virtual agent through online chat became feasible. My PhD project proposes to investigate how LLMs based virtual agents can be utilized to capture knowledge and priorities of a participant in MR-based immersive collaboration, creating a virtual surrogate which can interact with remote collaborators on behalf of a person who is not available at particular time.

Proposed Approach Our goal is to support a truly iterative process involving asynchronous actions made by two collaborators (A and B) at different times. I propose to use an LLM at two stages of this iterative process (see figure 1). First, the LLM will capture knowledge associated with *collaborator A*'s actions. To achieve this, the system will generate textual descriptions of *collaborator A*'s actions, which will be sent to the LLM. The LLM will then be able to collect this information and to ask questions about the rationale behind these actions. Second, the LLM will present later the information collected on *collaborator A*'s actions to the *collaborator B*. This presentation can take several forms: first, the LLM could present a summary of activities, or it could help the *collaborator B* to control the replay of the actions and explain the rationale behind these actions. These two steps will be repeated for each iteration. In this way, the LLM knowledge of the collaborative activities will be progressively expanded throughout the iterative process.

Application context I plan to test and assess the proposed approach on an asynchronous collaborative design scenario. This scenario will target interior design tasks between two experts: an interior designer and a lighting designer, within a virtual environment. The LLM will engage by asking relevant questions about prior design decisions or explaining the design rationale to the collaborators when needed. Later, the PhD work could be extended to a broader range of collaborative contexts and application domains.

First prototype At this stage of my PhD, I have developed a first prototype with Unity. The Meta SDK is used to provide text-to-speech and speech-to-text capabilities, enabling voice interaction and transcription. This transcription is then processed by Google Gemini which provides the summary of the user activity to the other collaborator. Additionally, the application incorporates functionality to store and manage modifications of virtual objects.

4 FUTURE RESEARCH

The forthcoming research will focus on: (i) Finding strategies to present insights from LLMs to the collaborators, such that the information is easy to grasp and the overview of previous activities can be displayed in an efficient way; (ii) Investigating the limitations of LLMs to support asynchronous collaboration, as an LLM

may raise bias and tend to produce verbose responses, which obscure the collaboration activities; and (iii) Designing the right tool to implement record-and-replay functionality in the virtual environment which should be able to record different modalities, such as hand and head movement, audio and object manipulation, and display the recordings in a cohesive manner.

5 CONCLUSION

This PhD project investigates how LLMs can improve asynchronous collaboration in MR. I would like to discuss the following research questions in the Doctoral Consortium: (i) How can the LLM capture more knowledge during the recording? (ii) How to represent the virtual agent in the virtual environment?

ACKNOWLEDGMENTS

This work was partially supported by European Union's Horizon Europe Research and Innovation Program under Grant n°101126644 (SEED). It is also supported by French government funding managed by the National Research Agency under the Investments for the Future program (PIA) grant ANR-21-ESRE-0030 (CONTINUUM) and as part of the France 2030 program, reference ANR-22-EXEN-0003 (PEPR eNSEMBLE / PILOT).

REFERENCES

- [1] H. Cho, B. Yuan, J. D. Hart, Z. Chang, J. Cao, E. Chang, and M. Billinghurst. Time travellers: An asynchronous cross reality collaborative system. In *IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*, pp. 848–853, 2023. doi: 10.1109/ISMAR-Adjunct60411.2023.00186 1
- [2] K. Chow, C. Coyiuto, C. Nguyen, and D. Yoon. Challenges and design considerations for multimodal asynchronous collaboration in vr. *Proceedings of the ACM on Human-Computer Interaction*, 3(CSCW), Nov. 2019. doi: 10.1145/3359142 1
- [3] A. R. Fender and C. Holz. Causality-preserving asynchronous reality. In *Proceedings of the Conference on Human Factors in Computing Systems*, CHI '22. ACM, 2022. doi: 10.1145/3491102.3501836 1
- [4] C. G. Fidalgo, Y. Yan, H. Cho, M. Sousa, D. Lindlbauer, and J. Jorge. A survey on remote assistance and training in mixed reality environments. *IEEE Transactions on Visualization and Computer Graphics*, 29(5):2291–2303, 2023. doi: 10.1109/TVCG.2023.3247081 1
- [5] A. Irlitti, R. T. Smith, S. Von Itzstein, M. Billinghurst, and B. H. Thomas. Challenges for asynchronous collaboration in augmented reality. In *IEEE Int. Symp. on Mixed and AR (ISMAR-Adjunct)*, pp. 31–35, 2016. doi: 10.1109/ISMAR-Adjunct.2016.0032 1
- [6] L. S. Lenz, A. R. Fender, J. Chatain, and C. Holz. Comparing synchronous and asynchronous task delivery in mixed reality environments. *IEEE Transactions on Visualization and Computer Graphics*, 30(5):2776–2784, 2024. doi: 10.1109/TVCG.2024.3372034 1
- [7] A. Mayer, T. Combe, J.-R. Chardonnet, and J. Ovtcharova. Asynchronous manual work in mixed reality remote collaboration. In *Extended Reality*, pp. 17–33. Springer, 2022. 1
- [8] A. Schäfer, G. Reis, and D. Stricker. A survey on synchronous augmented, virtual, and mixed reality remote collaboration systems. *ACM Computing Surveys*, 55(6), Dec. 2022. doi: 10.1145/3533376 1
- [9] M. Sereno, X. Wang, L. Besançon, M. J. McGuffin, and T. Isenberg. Collaborative work in augmented reality: A survey. *IEEE Transactions on Visualization and Computer Graphics*, 28(6):2530–2549, 2022. doi: 10.1109/TVCG.2020.3032761 1
- [10] S. Thanyadit, P. Punpongsanon, T. Piumsomboon, and T.-C. Pong. XR-LIVE: Enhancing asynchronous shared-space demonstrations with spatial-temporal assistive toolsets for effective learning in immersive virtual laboratories. *Proc. of the ACM on Human-Computer Interaction*, 6(CSCW1), Apr. 2022. doi: 10.1145/3512983 1
- [11] J. Zhang, B. Han, Z. Dong, R. Wen, G. A. Lee, S. Hoermann, W. Zhang, and T. Piumsomboon. Virtual triplets: A mixed modal synchronous and asynchronous collaboration with human-agent interaction in virtual reality. In *Extended Abstracts of the CHI Conference*, CHI EA '24. ACM, 2024. doi: 10.1145/3613905.3650766 1